

FLF Strategy Reset Cheatsheet

Simulation-Driven Experimental Design

Purpose: Decide the grand challenge, year-4 flagship outcome, MVP demonstrators, paper pathway, and impact plan.

0. Meeting frame

Time: 1 hour

Audience: Direct team

Mode: Decisive first, motivational second

Main output: A shared direction that connects software + OVal-DB into one engineering methodology.

1. Opening statement

We are 1 year 9 months into a 4-year Future Leaders Fellowship.

The original roadmap has evolved, which is a good thing. Today is not a normal status meeting. The purpose is to remap the fellowship around a single coherent goal:

Creating a simulation-driven experimental design platform, validated against real experimental data, that becomes a new engineering methodology for high-value component validation inside and outside fusion.

We have two main arms:

1. Software for experimental design and analysis

pyvale, Riley, synthetic diagnostics, DIC analysis, validation metrics, sensor placement, test design.

2. OVal-DB: the Open Validation Database

The real experimental datasets that prove the software works in practice.

By the end of this meeting, I want us to agree:

- The **grand challenge**
 - The **year-4 flagship outcome**
 - The **first MVP demonstrator**
 - The **key technical milestones**
 - The **paper and impact pathway**
 - Who owns what next
-

2. The north-star question

Ask the team:

By the end of year 4, what should be demonstrably true that is not true now?

Capture answers under three headings:

Software capability	Validation evidence	Adoption / impact
What can the tools do?	What does OVal-DB prove?	Who uses this and why?

Useful prompts:

- What should an external engineer be able to do with our methodology?
- What should OVal-DB prove?
- What should our software predict, optimise, or explain?
- What would make this clearly more than another DIC/simulation/data project?
- What would make people in fusion, aerospace, nuclear, or advanced manufacturing care?

3. Candidate grand challenge statements

Discuss and refine these.

Option A — methodology framing

Create a simulation-driven experimental design methodology that tells engineers what to measure, where to measure it, and how much trust to place in the resulting validation evidence.

Option B — software/data platform framing

Build and validate an open software-and-data platform for designing high-value validation experiments using predictive simulation, synthetic diagnostics, and real-world benchmark datasets.

Option C — impact framing

Move high-value component validation from expert-led experimental judgement to simulation-guided, uncertainty-aware, evidence-based engineering decision making.

Suggested combined version

Create a simulation-driven experimental design methodology that tells engineers what to measure, where to measure it, and how much trust to place in the resulting validation evidence, moving high-value component validation from expert

judgement toward uncertainty-aware, evidence-based engineering decision making.

Decision to make:

What is our preferred grand challenge statement?

4. Year-4 flagship outcome

Proposed strawman:

By the end of the first 4 years, we will have demonstrated a simulation-driven experimental design platform that can take a candidate validation problem, simulate plausible diagnostic outputs, optimise what should be measured, compare predictions against open benchmark experimental data, and produce uncertainty-aware validation evidence suitable for engineering decision making.

Short version:

A validated software-and-data workflow for deciding what to measure, where to measure it, and how to interpret the result for high-value component validation.

Decision to make:

Is this the right year-4 flagship outcome?
What would we add, remove, or sharpen?

5. Candidate MVP demonstrators

We cannot validate the whole methodology at once. We need MVP demonstrators that are small enough to publish, but strong enough to prove the platform.

MVP 1 — Synthetic-to-real image-based validation

Core question:

Can we use Riley + pyvale + real OVal-DB DIC data to separate synthetic-image error, DIC systematic error, and physical-model validation error?

Software side:

- Riley
- pyvale
- synthetic DIC image generation

- renderer convergence
- DIC uncertainty
- diagnostic simulation

OVal-DB side:

- Controlled DIC benchmark datasets
- Known geometry/loading/boundary conditions
- Real image data with uncertainty information

Why it matters:

This proves that our synthetic diagnostic workflow can support real image-based validation.

Possible paper:

Renderer-converged synthetic image generation for uncertainty-aware DIC validation.

MVP 2 — Sensor or image placement optimisation

Core question:

Can we use simulation to decide where sensors, cameras, or image regions should be placed to maximise validation value?

Software side:

- Sensor simulation
- Information gain
- Entropy / KL divergence
- Optimisation
- Mechanical and thermal sensor models
- DIC/image-region design

OVal-DB side:

- Real experiment with multiple possible sensor layouts
- Retrospective or prospective comparison of sensor choices
- Benchmark task: predict which measurement configuration is most informative

Why it matters:

This is easy for industry to understand:

Put the sensors where they generate the most useful validation evidence.

Possible paper:

Simulation-driven sensor placement for high-value component validation using open benchmark datasets.

MVP 3 — Validation metrics / model adequacy

Core question:

Given simulation predictions and experimental observations, can we quantify whether the model is good enough for the decision being made?

Software side:

- Validation metrics
- Uncertainty propagation
- Model discrepancy
- Decision thresholds
- Evidence scoring

OVal-DB side:

- Datasets with known loads, diagnostics, and uncertainty
- Comparison between model predictions and measurements
- Clear acceptance/rejection/adequacy criteria

Why it matters:

This turns experimental data into engineering evidence.

Possible paper:

Decision-oriented validation metrics for simulation-driven experimental design.

MVP 4 — MT2.0 / test design

Core question:

Can we design a test that produces the most useful validation evidence for a target model, component, or failure mode?

Software side:

- Test design
- Synthetic diagnostics
- Forward models
- Uncertainty-aware optimisation
- Comparison of candidate experiments

OVal-DB side:

- Physical experiments designed or reinterpreted through the platform
- Benchmark datasets for testing alternative experimental designs

Why it matters:

This is the broadest and most ambitious version of the methodology.

Possible paper:

Simulation-driven test design for model validation of high-value engineering components.

6. MVP selection discussion

Ask the team:

Which MVP gives us the best combination of publishability, strategic value, and impact?

Score each candidate quickly from 1–5.

MVP	Publishable soon?	Strategic value?	OVal-DB fit?	Industry/internal impact?	Total
Synthetic-to-real DIC validation					
Sensor/image placement optimisation					
Validation metrics / model adequacy					
MT2.0 / test design					

Decision to make:

What is our first MVP demonstrator?

7. Paper ladder

The paper pathway should tell one coherent story.

Paper 1 — Foundation capability

Theme: Synthetic diagnostics / renderer convergence / DIC uncertainty

Purpose:

Prove that the synthetic data pipeline is trustworthy.

Possible title:

Renderer-Converged Synthetic Image Generation for Uncertainty-Aware DIC Validation

Paper 2 — Experimental design method

Theme: Sensor placement, image placement, or validation metrics

Purpose:

Prove that the software can make useful experimental design decisions.

Possible title:

Simulation-Driven Sensor Placement for High-Value Component Validation

Paper 3 — End-to-end OVal-DB validation case study

Theme: Full software + real dataset demonstrator

Purpose:

Prove that the methodology works on real validation data.

Possible title:

An Open Benchmark Demonstration of Simulation-Driven Experimental Design for Engineering Validation

Paper 4 — Broader methodology / impact paper

Theme: The complete engineering methodology

Purpose:

Position the fellowship as a new approach to validation, not just a set of tools.

Possible title:

Simulation-Driven Experimental Design as an Engineering Methodology for High-Value Component Validation

8. Impact pathway

Use this pipeline:

Stage	Output	Purpose
MVP demo	Small working workflow	Show the concept works
Paper	Peer-reviewed method/proof	Establish credibility
OVal-DB release	Dataset + benchmark task	Let others test against it
Internal adoption	Apply to UKAEA/fusion programme	Prove operational value
External adoption	Transfer to non-fusion engineering	Show the method generalises

Ask:

- What is the first demonstrator we can show internally?
- What dataset supports it?
- What software needs to exist?
- Who is the first internal customer?
- Who outside fusion would care first?
- What would a convincing “demo day” version look like?
- What needs to be open, documented, packaged, or simplified for adoption?

9. Six-month milestones

Ask:

What are the three things that, if we deliver them in the next six months, make the year-4 outcome feel inevitable?

Possible milestone categories:

Software milestones

- Riley/pyvale synthetic-to-real DIC workflow
- Sensor simulation for mechanical and/or thermal diagnostics
- Initial sensor placement optimisation method
- Validation metric prototype
- Reproducible benchmark scripts
- Documentation and tutorial examples

OVal-DB milestones

- First curated benchmark dataset
- Dataset metadata standard
- Uncertainty description for measurements

- Baseline simulation model
- Baseline validation task
- Public or internal release candidate

Impact milestones

- First MVP paper drafted
- Internal demonstration to target programme
- External conference abstract
- OVal-DB landing page or technical note
- Industry-facing explainer
- Adoption target identified

Decision to make:

What are our top three six-month milestones?

10. Ownership

Before closing, assign owners.

Workstream	Owner	Next concrete output	Target date
Grand challenge wording			
MVP demonstrator			
Paper 1			
Paper 2			
OVal-DB first dataset			
Software workflow			
Impact / adoption plan			

11. Closing question

End with this:

What would make us proud to say, at the end of year 4, that this fellowship changed how engineering validation is done?

Then close with:

Our goal is not just to write software or collect datasets. Our goal is to create a new engineering methodology: simulation-driven experimental design for high-value component validation. The software and OVal-DB are the two halves of the same proof.

12. Final decisions to capture

By the end of the meeting, capture:

- ☐ Grand challenge statement
- ☐ Year-4 flagship outcome
- ☐ First MVP demonstrator
- ☐ Key Oval-DB dataset for MVP
- ☐ First software workflow to demonstrate
- ☐ Top 3 six-month milestones
- ☐ Paper ladder
- ☐ First internal impact target
- ☐ First external impact target
- ☐ Owners and next outputs

One-sentence summary

We are turning software and Oval-DB into one coherent engineering methodology for deciding what to measure, where to measure it, and how to interpret the resulting evidence for high-value component validation.